

captures the ATE in logs for those who would have a positive outcome regardless of their treatment status. The parameter $\theta_{\text{Intensive}}$ is scale-invariant but is not point-identified from the marginal distributions of the potential outcomes (thus avoiding the trilemma in [Proposition 2](#)), and therefore cannot be consistently estimated without further assumptions.²⁹ However, [Lee \(2009\)](#) popularized a method for obtaining bounds on $\theta_{\text{Intensive}}$ under the monotonicity assumption that, for example, everyone with positive earnings without receiving a training would also have positive earnings when receiving the training.³⁰ Bounds on $\theta_{\text{Intensive}}$ can be reported alongside measures of the extensive margin effect, such as the change in the probability of having a non-zero outcome, $P(Y(1) > 0) - P(Y(0) > 0)$. One can also potentially tighten the bounds (or restore point-identification) by imposing additional assumptions on the joint distribution of the potential outcomes—we provide an example of this in our application to [Carranza et al. \(2022\)](#) below; see [Zhang et al. \(2008, 2009\)](#) for related approaches.³¹

We note that the parameter $\theta_{\text{Intensive}}$ is generally distinct from the “intensive margin” marginal effects implied by two-part models (2PMs), which were recommended for scenarios with zero-valued outcomes by [Mullahy and Norton \(2023\)](#), among others. In [Appendix D](#), we consider the causal interpretation of the marginal effects of 2PMs, building on the discussion in [Angrist \(2001\)](#). Our decomposition shows that the marginal effects from 2PMs yield the sum of a causal parameter similar to $\theta_{\text{Intensive}}$ as well as a “selection term” comparing potential outcomes for individuals for whom treatment only has an intensive margin effect to those with an extensive margin effect. It thus will generally be difficult to ascribe a causal interpretation to the marginal effects of 2PMs without assumptions about this selection.

5 Empirical applications

In this section, we focus on three concrete empirical applications to illustrate how the alternative parameters described in [Section 4](#) can be estimated in practice. To illustrate a range of possible applications, we consider a randomized controlled trial, a difference-in-differences design, and an instrumental variables design.

5.1 An RCT setting: [Carranza et al. \(2022\)](#)

[Carranza et al. \(2022\)](#) conduct a randomized controlled trial (RCT) in South Africa. Individuals randomized to the treatment group are provided with certified test results that they can show to

²⁹ $\theta_{\text{Intensive}}$ also does not take the form $E_P[g(Y(1), Y(0))]$, although it can be written as

$$\frac{E_P[\mathbb{1}[Y(1) > 0, Y(0) > 0] \log(Y(1)/Y(0))]}{E_P[\mathbb{1}[Y(1) > 0, Y(0) > 0]]},$$

where both the numerator and denominator take this form.

³⁰See, also, [Zhang and Rubin \(2003\)](#) for related results, including bounds without the monotonicity assumption.

³¹We note that the [Lee \(2009\)](#) bounds will tend to be tight when the extensive margin effect is close to zero. As noted in [Remark 2](#), this is precisely the setting where ATEs for log-like transformations are relatively insensitive to finite changes in scale.

prospective employers to vouch for their skills. Individuals in the control group do not receive test results.³² They then investigate how this treatment impacts labor market outcomes such as employment, hours worked, and earnings. We focus here on the effects on hours worked.

Original specification and sensitivity to units. Carranza et al. (2022) estimate the effect of their randomized treatment on the inverse hyperbolic sine of weekly hours worked. Formally, they estimate the OLS regression specification

$$\text{arcsinh}(Y_i) = \beta_0 + D_i\beta_1 + X_i'\gamma + u_i, \quad (3)$$

where Y_i is average weekly hours worked for unit i , D_i is an indicator for whether unit i was in the treatment group, and X_i is a vector of controls.³³ Their estimate of the ATE ($\hat{\beta}_1$) is 0.201 (see column (1) in Table 3). They interpret this as a 20% change in hours: “Certification increases average weekly hours worked, coded as zero for nonemployed candidates, by 20 percent” (p. 3560).

	arcsinh(weekly hrs)	arcsinh(yearly hrs)	arcsinh(FTEs)
Treatment	0.201 (0.052)	0.417 (0.096)	0.031 (0.012)
Units of outcome:	Weekly Hrs	Yearly Hrs	FTEs

Table 3: Estimates using $\text{arcsinh}(Y)$ with different units of Y in Carranza et al. (2022)

Note: This table shows estimates of the average treatment effect in Carranza et al. (2022) on the inverse hyperbolic sine of hours worked, estimated using (3). In the first column, the outcome is the inverse hyperbolic sine of *weekly* hours, as in the original paper. The remaining columns use the inverse hyperbolic sine of annualized hours (weekly hours times 52) or the inverse hyperbolic sine of the number of full-time equivalents worked (weekly hours divided by 40). Standard errors are clustered at the assessment date (the unit of treatment assignment) as in the original paper.

However, the results in Section 2 suggest that the estimate of β_1 should not be interpreted as a percentage effect, since it depends on the units of the outcome. To illustrate this, in columns (2) and (3) we re-estimate specification (3) with Y_i redefined to be (a) yearly hours worked, i.e. weekly hours times 52, or (b) the number of full-time equivalents (FTE) worked, i.e. weekly hours divided by 40. The results change quite substantially depending on the units used, with an estimate of 0.417 using yearly hours and 0.031 using FTEs. We therefore turn next to alternative approaches with a percentage interpretation in this setting.

Percentage changes in the average. The average number of (weekly) hours worked was 9.84 in the treated group and 8.85 in the control group. A simple summary of the treatment effect is

³²Some individuals are also assigned to a “placebo” arm in which they are provided the test results but the form does not include the individual’s name, and thus cannot credibly be shared with employers. We focus on the effect of the main treatment relative to the pure control group.

³³Carranza et al. (2022) include individuals receiving the “placebo” treatment in the sample and add an indicator for receiving the placebo treatment in X_i . We follow the same practice, although the results are similar if units receiving the placebo treatment are dropped.

thus that average hours worked were 11% higher in the treated group ($9.84/8.85 = 1.11$). This is an estimate of the parameter $\theta_{\text{ATE}\%} = E[Y(1) - Y(0)]/E[Y(0)]$ discussed in [Section 4.1](#) above. A numerically equivalent way to obtain this estimate of 11% is to use Poisson quasi-maximum likelihood estimation (Poisson QMLE) to estimate

$$Y_i = \exp(\beta_0 + \beta_1 D_i) U_i \quad (4)$$

and then calculate $\hat{\theta}_{\text{ATE}\%} = \exp(\hat{\beta}_1) - 1 = 0.11$ (see column (1) in [Table 4](#)).³⁴ This formulation in terms of Poisson QMLE is useful since it allows us to include covariates to potentially increase precision. Column (2) of [Table 4](#) shows the estimate of $\hat{\theta}_{\text{ATE}\%}$ from estimating

$$Y_i = \exp(\beta_0 + \beta_1 D_i + X_i' \gamma) U_i \quad (5)$$

by Poisson QMLE, with smaller standard errors than in column (1) (0.069 vs. 0.081).

	(1)	(2)
β_0	2.180 (0.058)	0.150 (0.311)
β_1	0.106 (0.072)	0.150 (0.060)
Implied Prop. Effect	0.112 (0.081)	0.150 (0.069)
Covariates	N	Y

Table 4: Poisson Regression and Implied Proportional Effects in [Carranza et al. \(2022\)](#).

Note: the first two rows of column (1) show the estimates of the coefficients β_0 and β_1 in (4), estimated using Poisson QMLE. The third row shows the implied estimate of the proportional effect, $E[Y(1) - Y(0)]/E[Y(0)]$, calculated as $\hat{\theta}_{\text{ATE}\%} = \exp(\hat{\beta}_1) - 1$. The second column shows analogous estimates using (5), which adds controls for pre-treatment covariates (we do not show the coefficients on the controls in the interest of brevity). Standard errors are clustered at the assessment date (the unit of treatment assignment) as in the original paper.

Separate estimates for the extensive/intensive margins. As shown in [Table 1](#), the treatment in [Carranza et al. \(2022\)](#) has an estimated extensive margin treatment effect of 0.055, meaning that it increases the fraction of people with positive hours worked by 5.5 percentage points. We may be interested in whether the overall 11% increase in hours worked is driven entirely by the extensive margin, or whether there is an intensive margin effect. That is, does the treatment increase hours only by bringing people into the labor force, or does it also allow people who would have worked anyway to find jobs with more hours (e.g. full-time instead of part-time)? To this end, we can use the method of [Lee \(2009\)](#) to compute bounds for the effect of the treatment for “always-takers” who

³⁴This estimation is done in the sample of treated units and control units, discarding the placebo group. One could equivalently retain the units in the placebo group and add an indicator for the placebo group to (4).

would have positive hours worked regardless of treatment ($Y(1) > 0, Y(0) > 0$).³⁵ The Lee bounds approach requires the monotonicity assumption that anyone who would work positive hours without the treatment would also work positive hours when treated (i.e., $P(Y(1) = 0, Y(0) > 0) = 0$). This seems reasonable if workers only share the information provided by the treatment when it helps their job prospects. It could be violated, however, if workers mistakenly share their test score results when in fact employers view them negatively.

Column 1 of [Table 5](#) reports bounds of $[-0.20, 0.28]$ for the effect of the treatment on log hours worked by the always-takers, while Column 2 shows bounds of $[-6.67, 2.77]$ for weekly hours (in levels). Unfortunately, in this setting the Lee bounds are fairly wide, including both a zero intensive-margin effect as well as fairly large intensive-margin effects (up to 28 log points). Thus, without further assumptions, the data is not particularly informative about the size of the intensive margin.

We can, however, say more if we are willing to impose some assumptions about how the always-takers, who would work regardless of treatment status ($Y(1) > 0, Y(0) > 0$), compare to the compliers ($Y(1) > 0, Y(0) = 0$), who only work positive hours when receiving the treatment. We might reasonably expect that the compliers are negatively selected relative to the always-takers and thus would work fewer hours when receiving treatment. We can formalize this by imposing that $E[Y(1) | \text{Complier}] = (1 - c)E[Y(1) | \text{Always-taker}]$, i.e. that average hours worked for compliers under treatment is $100c\%$ lower than for always takers. Columns 3 through 5 of [Table 5](#) report estimates of the average effect on the always-takers, assuming $c = 0, 0.25$, and 0.5 , respectively.³⁶ If we assume that always-takers and compliers work an equal number of hours under treatment ($c = 0$), then our point estimates suggest that there is actually a negative intensive-margin effect for the always-takers (-1.02 weekly hours). Under the assumption that compliers work 25% fewer hours ($c = 0.75$), the estimated effect for always-takers is near zero (-0.07 weekly hours), consistent with no important intensive margin. Finally, if we assume compliers work half as many hours as the always-takers ($c = 0.5$), then our estimates suggest a positive intensive margin effect (0.95 weekly hours). Our assessment of the importance of the intensive margin thus depends on how negatively-selected we think compliers are relative to always-takers.

5.2 A DiD setting: [Sequeira \(2016\)](#)

[Sequeira \(2016\)](#) studies a decrease in tariffs on trade between Mozambique and South Africa which occurred in 2008. She is interested in whether the reduction in tariffs reduced bribes paid to customs officers (among other outcomes). To study this question, she utilizes a difference-in-differences design comparing the change in bribes paid for products that were affected by the tariff change to that for a comparison group of products that did not experience a change in tariffs.

³⁵We again exclude units receiving the “placebo treatment.”

³⁶Under the assumptions in [Lee \(2009\)](#), $E[Y(1) | Y(1) > 0] = \theta E[Y(1) | \text{Always-taker}] + (1 - \theta)E[Y(1) | \text{Complier}]$, where $\theta = P(Y(0) > 0) / P(Y(1) > 0)$. Plugging in $E[Y(1) | \text{Complier}] = (1 - c)E[Y(1) | \text{Always-taker}]$, it follows that $E[Y(1) | \text{Always-taker}] = 1 / (\theta + (1 - c)(1 - \theta)) E[Y(1) | Y(1) > 0]$. Further, $E[Y(0) | \text{Always-taker}] = E[Y(0) | Y(0) > 0]$. Our estimation plugs in sample analogs to these expressions to estimate $E[Y(1) - Y(0) | \text{Always-taker}]$.